XAIFOOLER Algorithm

Algorithm 1 Adversarial Explanation Generation

Input: **target model** f, **Original Document** d_o, **Base Explanation** e_b, **Maximum Perturbation Threshold** p_t, **Current Perturbations** p_c **Current Similarity** s
Output: **Perturbed Document** d_p, **Perturbed Explanation** e_p, **Updated Similarity** s
Initialize: $e_p \leftarrow e_b, i \leftarrow 0, s \leftarrow 1, p_c \leftarrow 0$
Initialize: I $\leftarrow$ indices of replacement candidates that satisfy C
while $I \neq \varnothing$ *and* $p_c < p_t$ **do**
    s_i = RBO(d_p, perturb($d_p[I[i]]$)
    if $s_i < s$ **then**
        $s \leftarrow s_i$
        $e_p \leftarrow$perturb($d_p[I[i]]$)
    end if
    $I[i] = \varnothing$
    $i \leftarrow i + 1$
end while
return (d_p, e_p, s)

Effect of Parameter p

$$p \to 0$$

focus on top-ranked features.

$$p \to 1$$

consider deeper ranks.(For depth D, $E[D] = \frac{1}{1-p}$)
Allows tuning importance of top explanation features.

Results

XAIFOOLER significantly manipulates explanations.
Findings:

- larger ranking shifts than baselines
- predictions remain unchanged
- semantic similarity preserved

Implication:
LIME explanations are vulnerable to adversarial manipulation.

Limitations

Several limitations remain:

- Only evaluates LIME (not SHAP or other XAI)
- Greedy algorithm may not find optimal perturbations
- Computational cost is high
- Dependence on surrogate model quality

Future Research Directions

Possible extensions:

- source of instability within LIME
- stability analysis of SHAP explanations
- theoretical bounds on explanation robustness
- defense mechanisms against explanation attacks
- robust explanation frameworks